

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING TECHNOLOGY
END OF 1ST & 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAM: HND I Electrical and Electronics Engineering Technology.

UNIT: 2

COURSE TITLE: Electrical Material Science.

COURSE CODE: EEC315

TIME ALLOWED: 2 ½ Hours

INSTRUCTION: Attempt any Four (4) Questions only. All question carries equal marks. Do not write on the question paper

QUESTION ONE

- (a) Derive an expression to show the Neil Bohr's assumption that the centrifugal force on the orbit must be equal to columbic force between it and the electron; if centrifugal force = MV^2/r . (18mks)
- (b) States five (5) transitional jumps specified by Neil Bohr? (3mks)
- (c) Briefly explain Bohr's theory of the hydrogen atom. (4mks)

QUESTION TWO

- (a) Differentiate between structure of the atomic level and the properties of an atom. (13mks)
- (b) States three (3) expected benefits of an Engineer from material science? (3mks)
- (c) States four (4) postulations of Neil Bohr? (9mks)

QUESTION THREE

- (a) State the principle of the Thompson Atom model (Rutherford theory). (6mks)
 - (b) With the aid of diagram show the quantum jumps specified by Bohr. (3mks)
 - (c) Find the energy of an electron at rest. (4mks)
 - (d) A proton has 1.67×10^{-27} kg. Determine the energy of the proton at rest in the following:
 - (i) joules (3mks)
 - (ii) Mev. (3mks)
- If the proton is moving with velocity 0.5C, Find;
- (a) It's kinetic energy. (3mks)
 - (b) It's total energy in Mev. (3mks)

QUESTION FOUR

- (a) Briefly explain the following:
 - (i) Spectral lines. (7mks)
 - (ii) Photo-electric emission. (3mks)
 - (iii) Electron transmission from one energy level to another. (3mks)
- (b) Calculate the shortest wavelength in the pfund series of the hydrogen spectrum given that Rydbery constant $R = 1.09 \times 10^7 \text{ m}^{-1}$ also calculate the longest wave length. (12mks)

QUESTION FIVE

- (a) State the four (4) summarized principles postulated by Neil Bohr. (8mks)
- (b) Briefly explain photo-electric effect according to Albert Einstein. (5mks)
- (c) State the five (5) notable points with respect to photo electric effect. (12mks)

QUESTION SIX

- (a) Briefly explain the uncertainty principle of Heisenberg. (3mks)
- (b) The position and momentum of an electron is simultaneously determined. If the uncertainty in the position is 10^{-10} m , determine the uncertainty in the momentum. (3mks)
- (c) Find the uncertainty in the position of an electron that has an uncertainty speed of 10 ms^{-1} . (3mks)
- (d) If β of an excited speed of an electron is 128 ms^{-1} measured to an accuracy of 0.02%. find the uncertainty in position measurement. (8mks)
- (e) Briefly explain the following:
 - (i) Conductors. (2mks)
 - (ii) Insulators. (2mks)
 - (iii) Semiconductors. (2mks)
 - (iv) X-Ray diffraction. (2mks)

THE FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING TECHNOLOGY
END OF 1ST & 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAM: HND I Electrical and Electronics Engineering Technology.
COURSE TITLE: Analogue Electronics III.
COURSE CODE: EEC 314

UNIT: 3

TIME ALLOWED: 3 Hours

INSTRUCTION: Attempt any Five (5) Questions only. All question carries equal marks. Do not write on the question paper

- Q1. (a) The following quantities are measured in a CE amplifier circuit:
- With output a.c. short-circuited (i.e. $V_{ce} = 0$)
 $I_b = 10 \mu A$; $I_c = 1 \text{ mA}$; $V_{be} = 10 \text{ mV}$
 - With input a.c. open-circuited (i.e. $I_b = 0$)
 $V_{be} = 0.65 \text{ mV}$; $I_c = 60 \mu A$; $V_{ce} = 1 \text{ V}$
- Determine all the four h parameters.

- (b) Draw the equivalent h-parameters circuit for a two-port network of fig 1. (b) (14mks)

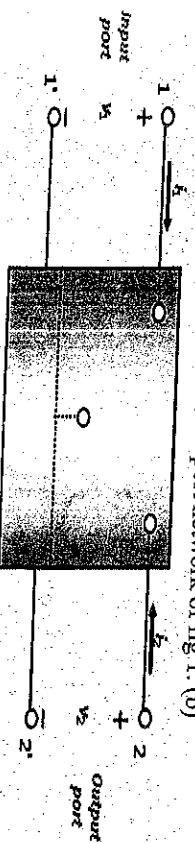


Figure 1b

- Q2. (a) List (without sketches) the various modes (configurations) of a differential amplifier. (4mks)
- (b) The transistor circuit of figure (2) has the following parameters; $h_{ie} = 1 \text{ k}\Omega$, $h_{fe} = 50$, $V_{cc} = +20 \text{ V}$, $I_{be} = 5 \times 10^{-4}$, $h_{oe} = 2 \times 10^{-5}$ Siemens, $R_b = 300 \text{ k}\Omega$, $R_c = 10 \text{ k}\Omega$ and $R_L = 10 \text{ k}\Omega$.
- Draw the equivalent h-parameters circuit. (5mks)
 - Determine the actual; I_{in} , I_{out} , A_i , and A_v . (11mks)

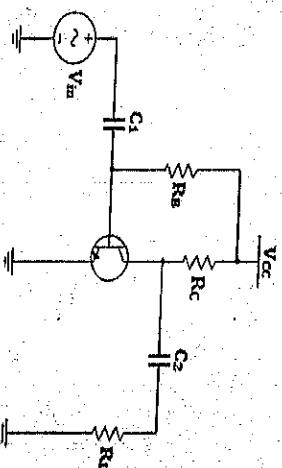


Figure 2b

- Q3. (a) List four (4) applications of the Silicon Control Rectifier (SCR). (6mks)
- (b) A forward breakdown Voltage of 115 V is applied to an SCR half-wave rectifier when the gate current of 1 mA flows. If a sinusoidal Voltage of 460V is applied, determine the following;
- Firing angle (4mks)
 - Conduction angle. (3mks)
 - Average Voltage. (4mks)
 - Average current for a load resistor of 200 Ω . (3mks)
- Q4. (a) Using a suitable diagram, differentiate between Triac and Diac in term of;
- Construction. (6mks)
 - Schematic symbol. (5mks)
- (b) A differential amplifier has the following parameters; $R_{c1} = R_{c2} = 21 \text{ k}\Omega$, $R_{b1} = R_{b2} = 1 \text{ k}\Omega$, $h_{ie1} = h_{ie2} = 14 \text{ k}\Omega$, $R_e = 20 \text{ k}\Omega$ and $h_{fe1} = h_{fe2} = 40$. Determine;
- Differential gain (A_d) (3mks)

- ii. Common mode gain (A_c) (3mks)
- iii. Common Mode Rejection Ratio (CMRR). (3mks)

Q5. (a) Design a three (3) input inverting summer using operational amplifier with gain for individual input equal to -1, -2, and -5 respectively. (12mks)

(b) For the operational amplifier of figure 3, $R_1 = R_2 = 5k\Omega$, $R_f = R_3 = 47k\Omega$, $V_1 = 70mV$, and $V_2 = 120mV$. Determine the value of V_{out} . (8mks)

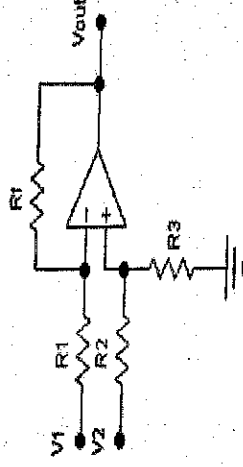


Figure 3b

Q6. (a) The following specifications are given for the differential amplifier of figure 4; $R_{C1} = R_{C2} = 2.5k\Omega$, $R_{B1} = R_{B2} = 70\Omega$, $R_E = 5k\Omega$, $+V_{CC} = 12V$, $-V_{EE} = -12V$, $V_{BE} = 0.7V$ and $\beta = 100$. Determine, (7mks)

- i. DC operating points I_{CQ} and V_{CEQ} (3mks)
- ii. Balanced output Voltage. (3mks)

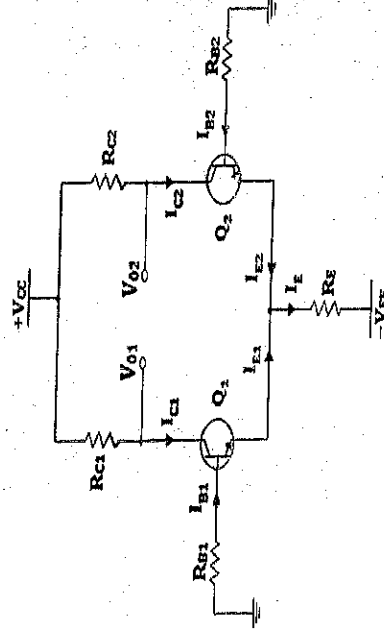


Figure 4

(b) From the areas of application of the operational amplifier listed below, design and briefly explain the basic principle of operation of only one out of the three. (10mks)

- 1) Voltage Limiters
- 2) Voltage Comparators
- 3) oscillators

Q7. (a) Enumerate four (4) types of noise and briefly explain two (2). (10mks)

(b) Given two uncorrelated noise sources that have $V_{n1}(rms) = 3\mu V$ and $V_{n2}(rms) = 7\mu V$. Calculate their total output rms value when combined. If it is required to maintain the total rms value at $6\mu V$, how much should $V_{n2}(rms)$ be reduced while $V_{n1}(rms)$ remains unchanged. (10mks)

THE FEDERAL POLYTECHNIC, MUBI
SCHOOL OF HUMANITIES AND CREATIVE ARTS TECHNOLOGY
DEPARTMENT OF LANGUAGES AND COMMUNICATION
1st AND 3rd SEMESTER MAIN EXAMINATION, 2021/2022 ACADEMIC SESSION

PROGRAMMES: HND I Bio/Microbiology Science Tech; Physic, Chemical Science; Mechanical Engineering; Animal Health, Animal Production, Crop Production, Agric Extension and Mgt; Civil Engineering, Public Admin; Lib. & Information Science Tech; Electrical Engineering, Computer Engineering, Agric. & Bio-Environmental; Mass Communication, Food Science, Nutrition & Diet, Civil Engineering.

COURSE TITLE: Use of English III
TIME ALLOWED: 2hrs 30mins

COURSE CODE: GNS 301

Instruction: Answer any four (4) questions.

Question One (1)

- a. Define sentence
- b. i. Identify the four types of sentence by function
ii. Give one example for each
- c. Read the following sentences and choose the correct option from the words in the brackets.
 - i. Neither Hussaini nor Jonah (is/are) eating.
 - ii. Amina and Hauwa were (both/all) students of the Federal Polytechnic, Mubi.
 - iii. Either You or Stella (is/arc) around.
 - iv. Fred is (tall/taller) than John.
 - v. One hundred naira (pay/pays) for a bottle of CocaCola.
 - vi. Aisha (give/gives) Mohammed food every day.
 - vii. Politics (arc/is) bitter.
 - viii. The lecturers are in (them/their) offices.

Question Two (2)

- a. Explain the term logical thinking.
- b. Differentiate between fact and opinion.
- c. Below are sentences C.i – v, identify the following sentences as either fact or opinion.
 - i. God loves everybody.
 - ii. All lecturers are punctual for their classes.
 - iii. The Federal Polytechnic, Mubi is an academic institution.
 - iv. Donald Trump of the United States of America is the best president in the world.

Question Three (3)

- a. What is an essay?
- b. Mention the four types of essay.
- c. Mention three characteristics of any two of the essay types.

Question Four (4)

- a. What is literature?
- b. List and explain the three genres of literature.
- c. Write notes on any three of the following literary terminologies.
 - i. Stanza
 - ii. Rhythm

- iii. Binjambment
- iv. Sonnet
- v. Theme

Question Five (5)

- a. What is concord?
- b. Identify the structure of the following sentences: Compound-complex, compound, simple, complex.
 - i. Sulaiman went to the market and Hauwa stayed at home to cook.
 - ii. She dances.
 - iii. When I was sleeping, he came.
 - iv. While I was cooking, he was still playing games.
 - v. When I grow up, I want to be a musician and my mother is proud of me.
- c. Identify the underlined part of the sentences given below as either subject or predicate.
 - i. The Federal Polytechnic Mubi, Adamawa State is an excellent academic institution.
 - ii. They have finished eating.
 - iii. Jones and Yusuf like watching football.
 - iv. English Language is spoken in Nigeria.
 - v. Nigeria needs divine intervention.

Question Six (6)

Write an essay of three pages on the topic: "If I had known, I would have got my Permanent Voter's Card (PVC)."

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END OF 1ST & 3RD SEMESTER EXAMINATION 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAMME: HND I Electrical and Electronics Engineering Technology

UNIT: 3

COURSE TITLE: Computer Packages

COURSE CODE: ICT 302

TIME ALLOWED: 3 Hours

INSTRUCTION: Attempt any Five (5) Questions only. All questions carry equal marks. Do not write on the question paper

QUESTION ONE

- (a) Define Database and state three (3) examples of database. (6 marks)
- (b) State four (4) advantages of electronic filing database over the manual filing. (8 marks)
- (c) Differentiate between a file processing system and a database management system. (6 marks)

QUESTION TWO

- (a) List and explain briefly any two (2) Access objects. (9 marks)
- (b) Highlight the general steps to construct a database. (7 marks)
- (c) Enumerate two advantages of database table with primary key over the one without. (4 marks)

QUESTION THREE

- (a) Define the term rendering as it relates to computer graphics. (4 marks)
- (b) Briefly explain the following types of rendering:
 - i. Volume rendering
 - ii. Surface rendering
- (c) Write short notes on the following:
 - i. Clipping
 - ii. Rasterization

QUESTION FOUR

- (a) Briefly describe the components of database management system (DBMS). (8 marks)
- (b) Given the following Fields; Name, Date of admission, Admission Number, Registration Number, Tuition fee, image. In tabular form state the data type to be assigned to each field above. (9 marks)
- (c) Differentiate between desktop DBMS and the Server DBMS. (3 marks)

QUESTION FIVE

- (a) Differentiate between Menu-driven system and Command-driven system. (8 marks)
- (b) Briefly explain the following types of plotters: Drum plotter, Flatbed plotter and Grid wheel plotter. (6 marks)
- (c) State the types of Coordinates systems with the general representation. (6 marks)

QUESTION SIX

- (a) List the common drawing aids used in found in CAD and explain briefly any two. (3 marks)
- (b) Write short notes on the following types of editing commands: Move Command, Mirror Command, Rotate Command and Erase Command. (8 marks)
- (c) Differentiate between entities and modifiers (3 marks)

QUESTION SEVEN

- (a) Write briefly on the following components of AUTO CAD software: Title Bar, Menu Bar, Ribbon, Command Window, and Command Line. (10 marks)
- (b) State the factors to be considered in choosing the software for a CAD drawing. (3 marks)
- (c) List any seven input devices used in CAD system. (7 marks)

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END OF 1ST & 3RD SEMESTER EXAMINATION, 2021/2022 ACADEMIC SESSION
SEPTEMBER, 2022.

PROGRAM: HND I Electrical and Electronics Engineering Technology.
COURSE TITLE: Electrical Circuit Theory III.
COURSE CODE: EEC 313
TIME ALLOWED: 2 ½ Hours
INSTRUCTION: Attempt any Five (5) Questions only. All question carries equal marks. Do not write on

UNIT: 3

the question paper

QUESTION ONE

- a) Write short notes on the phenomenon of transients as a response which may finally decay after a short period of time, taking into consideration the transient analysis, transient state and steady states. (12 mks)
- b) List the major variables that are present in AC/DC circuits. (4 mks)
- c) Transient occur as a result at least one of the four factors responsible. List the four factors (4 mks)

QUESTION TWO

An electromagnet connected to a 120 Vdc supply has resistance and inductance of 8 Ω and 4 H respectively. Determine:

- i. The maximum value of current flowing in the circuit. (3 mks)
- ii. Time constant of the circuit. (3 mks)
- iii. The value of induced e.m.f after a time equal to time constant of the circuit. (3mks)
- iv. The time for the current to rise to 70% of its final value. (4mks)
- v. The value of current in the circuit after 0.5 secs. (4mks)
- vi. The value of the current in the circuit 2.0 secs after disconnecting the supply. (3mks)

QUESTION THREE

- a) Define the term “transmission line” with respect to electrical power system and give two examples. (4 mks)
- b) Define the Wavelength of propagation with respect to transmission line: (4 mks)
- c) Why is resistance and conductance of a transmission line referred to as imperfection of conduction and insulation respectively? (8 mks)
- d) List the four (4) major parameters which must be considered while designing an electrical transmission line. (4 mks)

QUESTION FOUR

- a) A 6 km long transmission line has the following secondary line constants; Characteristics impedance, $Z_0 = (20\angle -22^\circ)\Omega$, Attenuation coefficient, $\alpha = 0.5\text{Np/km}$, Phase shift Coefficient, $\beta = 0.2\text{rad/km}$. Determine the **magnitude and the phase angle of the current** at the receiving end of the voltage, if the sending end voltage is $(10.0\angle 0^\circ)\text{V}_{rms}$. (12 mks)

- b) Copy and complete the table for “**laws of complex number operation**” below, given Z_1 , Z_2 and Z_3 as:

$$Z = x + jy, r\angle\theta$$

$$Z_1 = x_1 + jy_1 = r_1\angle\theta_1$$

$$Z_2 = x_2 + jy_2 = r_2\angle\theta_2$$

Addition	$Z_1 + Z_2$	?
Subtraction	$Z_1 - Z_2$	?
Multiplication	$Z_1 Z_2$	?
Division	Z_1 / Z_2	?

QUESTION FIVE

Define the following terms:

- | | | | |
|-------------------------------------|--------|-----------------------------------|--------|
| a.) Transfer function | (4mrk) | d.) Root Loci | (3mks) |
| b.) Frequency response of a circuit | (4mrk) | e.) Poles of a transfer function | (3mks) |
| c.) Bode plot | (3mrk) | f.) Zeroes of a transfer function | (3mks) |

QUESTION SIX

State the eight (8) rules that must be adhered in constructing root loci.

(20mks)

QUESTION SEVEN

- a) Using KVL, show that the equation of the transfer function of the circuit shown in Fig. below is given by:

$$H(\omega) = \frac{1}{j\omega C(R + j\omega L) + 1} \text{ in frequency domain}$$

(12 mks)

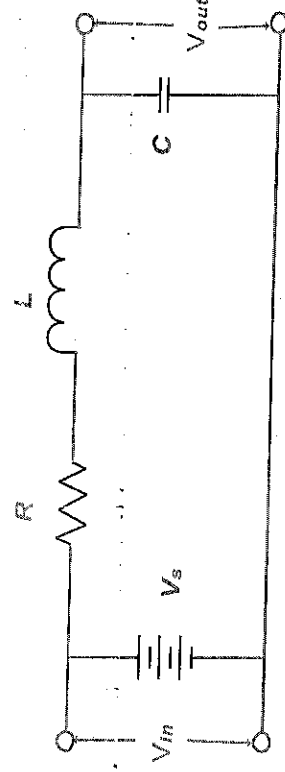


Fig. 3.0: Question Seven (7)

- b) State the four (4) possible transfer functions that can be possible in a system and also express the equations representing each.

(8 mks)

FEDERAL POLYTECHNIC MUBI
DEPARTMENT OF ELECTRICAL/ELECTRONICS ENGINEERING TECHNOLOGY
END OF 1ST & 3RD SEMESTER EXAMINATION 2021/2022 SESSION
SEPTEMBER, 2022

PROGRAMME:

HND I Electrical and Electronics Engineering

UNITS 2

COURSE TITLE:

ENGINEER IN SOCIETY

COURSE CODE:

EEEC 319

TIME ALLOWED:

2½ hrs

INSTRUCTION:

Answer any Four (4) questions. All question carry equal marks

QUESTION ONE

- (a) Briefly discuss how we can build and develop Engineers with demonstrable knowledge in engineering [12 marks]
- (b) Discuss briefly the three ways an Engineer transform today's technology for tomorrow's world ? {8 marks}

QUESTION TWO

- (a) Discuss briefly the Engineer's role from a specialist perspective [12 marks]
- (b) Briefly discuss the place of a wide range of engineering expertise, purpose built facilities and strong commercial links in National Development? [8 marks]

QUESTION THREE

- (a) Briefly discuss how as a consulting Engineer advising the Presidential Task Force (PTF) on COVID -19 guide them on the managing of social, educational, political and economic activities in a 'new normal' amid the persistent threat of COVID – 19 by technology – based and non pharmaceutical solutions.
- (b) State clearly how these solutions can encourage sustainable environment, mitigate identified risks, and promote good hygiene for non medicals and medicals.

QUESTION FOUR

- (a) Define the term civilization [2 marks]
- (ii) State three elements of civilization [3 marks]
- (b) (i) What do you understand by term anthropologist? [2 marks]
- (ii) List any five types of engineering structures you know [5 marks]
- (c) What are the social roles of engineers in the society? [8 marks]

QUESTION FIVE

- (a) State five areas of specialization each in the following engineering disciplines: [5 marks]
 - (i) Electrical and electronic engineering technology [5 marks]
 - (ii) Chemical engineering [5 marks]
 - (iii) Mechanical engineering [5 marks]
 - (iv) Inter disciplinary engineering [5 marks]

QUESTION SIX

- (a) Write short notes on the history of engineering in Nigeria [5 marks]
- (b) (i) Define professionalism [2 marks]
- (ii) State four qualities of professionalism [4 marks]
- (c) Outline the moral roles of engineers in the society [7 marks]
- (d) Identify the scientist and year of the following inventions; [1 mark]
 - (i) The first battery [1 mark]
 - (ii) The steam engine [1 mark]

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THE FEDERAL POLYTECHNIC MUBI

DEPARTMENT OF MECHANICAL ENGINEERING

1ST SEMESTER EXAMINATION 2021/2022 ACADEMIC SESSION

COURSE CODE/TITLE: EEI 311 ELECTRICAL MEASUREMENTS AND INSTRUMENTATION

PROGRAMME: HND I ELECTRICAL/ ELECTRONICS ENGINEERING

TIME ALLOWED: 3 HOURS

INSTRUCTION: ATTEMPT ANY FIVE QUESTION

QUESTION 1:

- a) What is measurement? (5 MARKS)
- b) All measurement systems consist essentially of three main parts; what are they? (12 MARKS)
- c) Define calibration (3MARKS)

QUESTION 2:

- a) Define the following i) repeatability ii) reproducibility iii) sensitivity iv) precision v) resolution vi) threshold (9 MARKS)
- b) Describe the principle, construction and working of thermistors as temperature measuring device (8 MARKS)
- c) State three advantages and three disadvantages of thermistors (3 MARKS)

QUESTION 3:

- a) What are the three ways in which instrumentation can be used for making measurement? (9 MARKS)
- b) Describe with the aid of a sketch, the principle, construction and working of thermocouples as temperature measuring devices (8 MARKS)
- c) State three advantages and three disadvantages of thermocouples (3 MARKS)

QUESTION 4:

- a) You are required to design a pressure measuring system that can withstand a lot of vibration and shock, is extremely sensitive, has a good frequency response and can give rapid response to change in pressure, which of the pressure measuring instruments will you use? (2 MARKS)
- b) With the aid of a diagram explain the principle, construction and working of the instrument mentioned in 2a above. (15 MARKS)
- c) What are the disadvantages of the instrument? (3 MARKS)

QUESTION 5:

- a) List the various methods used for inferential flow measurement. (8 MARKS)

- b) What are the advantages and disadvantages of variable head flow measuring instruments. (5 MARKS)
- c) The periodic maintenance of pressure measuring instruments consists of three things, what are they? (7 MARKS)

QUESTION 6:

- a) List the hydrostatic pressure methods used for liquid level measurement. (4 MARKS)
- b) Describe with the aid of a diagram the capacitance level indicator. (10 MARKS)
- c) Write down the advantages and the disadvantages of the capacitive level indicator. (6 MARKS)

QUESTION 7:

- a) Briefly describe the construction and working of a radiation level detector. (12 MARKS)
- b) What are the advantages and disadvantages of radiation level detector (8 MARKS)

THE FEDERAL POLYTECHNIC, MUBI

(School of Science and Technology)

Department of Mathematics and Statistics
End of 1st and 3rd Semester Examinations 2021/2022 Session

Course Code: MTH 311

Course Title: Advanced Algebra

Instruction: Answer any five (5) Questions

Time: 3 Hours

Question 1

- (a) Find the values of; (i) $\sinh^{-1} 1.475$ (ii) $\tanh^{-1} 0.75$
- (b) (i) Solve the equation, $12\cosh^2 x + 7\sinh x = 24$ for real values of x .
(ii) Prove that $1 - \tanh^2 x = \operatorname{sech}^2 x$

Question 2

- (a) (i) Explain the terms convergence and divergence of a power series.
(ii) Test the convergence or divergence of the series; $\frac{1}{5} + \frac{2}{6} + \frac{2^2}{7} + \frac{2^3}{8} + \frac{2^4}{9} + \dots$
- (b) (i) State the Taylor's theorem and use it to expand $\sin\left(\frac{\pi}{6} + h\right)$ in ascending powers of h upto the term h^4 .
(ii) Use Maclaurin's theorem to expand $\sinh x$ in ascending powers of x upto the fourth non-zero terms.

Question 3

- (a) Determine: (i) $\lim_{x \rightarrow 0} \left\{ \frac{x^2 - \sin 3x}{x^2 + 4x} \right\}$ (ii) $\lim_{x \rightarrow 0} \left\{ \frac{x \cos x - \sin x}{x^3} \right\}$
- (b) Find the range of values of x for which the following series is absolutely convergent:

$$x - \frac{x^2}{3 \times 5^2} + \frac{x^3}{4 \times 5^3} - \frac{x^4}{5 \times 5^4} + \frac{x^5}{6 \times 5^5} - \dots$$

Question 4

- (a) Let $A = \begin{pmatrix} 2 & -5 & 1 \\ 2 & 0 & -4 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -2 & -3 \\ 0 & -1 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 1 \\ 1 & -1 & -2 \end{pmatrix}$. Find $3A + 4B - 2C$.
If $A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{pmatrix}$, find (i) AA^t (ii) $A^t A$ (iii) what can you deduce from the results of (i) and (ii)?
- (b) Solve the set of equations using inverse:
 $x_1 + 2x_2 + x_3 = 4$
 $3x_1 - 4x_2 - 2x_3 = 2$
 $5x_1 + 3x_2 + 5x_3 = -1$

Question 5

- (a) (i) Prove by induction that $x^n - y^n$ has $x - y$ as a factor for all positive integer n .
(ii) Evaluate $\iint_R (x^2 + y^2) dx dy$ where R is the region between the line $y = x$ and the curve $y = x^2$.

- (b) Evaluate the following integrals: (i) $\int_0^2 \int_0^3 \int_0^1 (p^2 + q^2 - r^2) dp dq dr$ (ii) $\int_{-1}^3 \int_0^1 \int_0^2 (x + 2y - z) dx dy dz$

Question 6

- (a) If $\phi = x^2 yz^3$ and $\vec{A} = xz \vec{i} - y^2 \vec{j} + 2x^2 y \vec{k}$, find (i) $\nabla \phi$ (ii) $\nabla \cdot \vec{A}$ (iii) $\nabla \times \vec{A}$
- (b) (i) State the Green's theorem.
(ii) Verify the theorem in plane for $\oint_C (2xy - x^2) dx + (x + y^2) dy$ where C is the closed curve bounding the region between $y = x^2$ and $y^2 = x$.





THE FEDERAL POLYTECHNIC, MUBI.

**END OF FIRST SEMESTER 2021/2022 ACADEMIC SESSION EXAMINATIONS
PROGRAMME: HND I ELECTRICAL ENGINEERING**

COURSE COD: SDV 210

COURSE TITLE: INTRODUCTION TO ENTREPRENEURSHIP.

TIME ALLOWED: TWO HOURS THIRTY MINUTES

INSTRUCTION: ANSWER QUESTION ANY FIVE QUESTIONS

Q1 With a good example define Entrepreneur, Enterprise

(b) What is Entrepreneurship?

Q2 Your cousin whom stays in the village is seeking for advice on the business he wishes to venture into but his mean fears are failure, tell him why business suffers failures

Q3 List and explain the characteristics of Entrepreneurs

Q4 Explain the following

- i. Selection of business opportunity
- ii. Combination of factors of production
- iii. Establishment of business enterprise
- iv. Management of business Enterprise
- v. Decision making

Q5 Define Education, Training and Development

Q6 Write short note on the following

- i. Nigerian Customary law
- ii. Nigerian statutory law
- iii. Judicial precedent

Q7 Discuss Offer, Acceptance and Consideration

GOODLUCK

